

Magnetic transport and dark-photon acceptance in HPS

HPS-GPR v5.7.1

13 September 2026

The opening angles of an $A' \rightarrow e^+e^-$ decay decrease as its laboratory energy increases. This shifts the mass scale at which the daughter tracks enter a forward spectrometer. The calculation below connects that boost to the active silicon geometry and magnetic fields for representative 2015, 2016 and 2021 HPS configurations. The electron and positron are propagated through the field maps and tested against finite sensor faces using the run-dependent hit requirements. This replaces the assumed 70 mrad outer angle of v5.7.0 with a calculation of curved tracks through the detector.

The resulting fractions describe prompt decays at a fixed parent energy and direction. They include magnetic transport and the hit component of the selection; material interactions, hit finding, trigger, reconstruction and other analysis cuts remain outside the calculation. Zero-field curves are retained to show the effect of bending. The equations and detector assumptions precede the figures so this section can be inserted into a longer analysis note.

1 Two-body decay and laboratory acceptance

The beam energy controls the angular separation of the electron and positron from a boosted dark photon. We first consider a prompt $A' \rightarrow e^+e^-$ decay at the target, with the parent directed along the beam axis z . In units with $c = 1$, let m denote the parent mass and write its laboratory energy as $E_A = xE_{\text{beam}}$, where x is the fraction of the beam energy carried by the parent. At fixed m and x , the boost and daughter rest-frame momentum are

$$\gamma = \frac{E_A}{m}, \quad \beta = \sqrt{1 - \frac{m^2}{E_A^2}}, \quad E_* = \frac{m}{2}, \quad p_* = \sqrt{\frac{m^2}{4} - m_e^2}, \quad (1)$$

with $2m_e \leq m < E_A$. Define $c_* = \cos \theta_*$ and let ϕ_* be the rest-frame azimuth measured from the horizontal x axis. The two rest-frame four-vectors and their laboratory components are

$$p_{\pm}^{*\mu} = \left(\frac{m}{2}, \pm p_* \sqrt{1 - c_*^2} \cos \phi_*, \pm p_* \sqrt{1 - c_*^2} \sin \phi_*, \pm p_* c_* \right), \quad (2)$$

$$E_{\pm} = \gamma \left(\frac{m}{2} \pm \beta p_* c_* \right), \quad p_{z,\pm} = \gamma \left(\frac{\beta m}{2} \pm p_* c_* \right), \quad (3)$$

$$p_{x,\pm} = \pm p_* \sqrt{1 - c_*^2} \cos \phi_*, \quad p_{y,\pm} = \pm p_* \sqrt{1 - c_*^2} \sin \phi_*. \quad (4)$$

The signs label the opposite daughter directions. For a forward daughter, its polar angle and vertical projected angle are respectively

$$\tan \theta_{\pm} = \frac{p_* \sqrt{1 - c_*^2}}{p_{z,\pm}}, \quad \tan \theta_{y,\pm} = \frac{p_{y,\pm}}{p_{z,\pm}}. \quad (5)$$

The energy fractions are $E_{\pm}/E_A = (1 \pm \beta \rho c_*)/2$, where $\rho = \sqrt{1 - 4m_e^2/m^2}$. They approach the momentum fractions only when the daughters are ultrarelativistic. Thus fixing x fixes the pair energy, while the decay angle still determines how that energy is shared.

1.1 Equal-sharing mass scales

For $c_* = 0$, both daughters have energy $E_A/2$ and the same polar angle θ ; their opening angle is 2θ . The exact invariant-mass relation is

$$m^2 = E_A^2 \sin^2 \theta + 4m_e^2 \cos^2 \theta \quad \longrightarrow \quad m \simeq xE_{\text{beam}} \sin \theta \simeq xE_{\text{beam}} \theta. \quad (6)$$

The first approximation neglects the electron mass and the second assumes a small angle. In the vertical decay plane, $|\sin \phi_*| = 1$, this gives the mass scale associated with a specified vertical angle. For a general decay, the small-angle, relativistic relation is instead $m^2 \simeq E_+ E_- \theta_{+-}^2$, with θ_{+-} the daughter opening angle. Unequal sharing therefore changes the correspondence between a mass and either single-track angle.

A vertical beam gap excludes tracks close to the beam *plane*; a polar cone excludes tracks close to the beam *axis*. Their acceptances differ because $|\tan \theta_y| = \tan \theta |\sin \phi_*|$. In particular, an outer vertical edge gives an upper equal-sharing mass scale only in the specified vertical plane. With unrestricted azimuth it does not, by itself, give an upper mass cutoff. Horizontal sensor boundaries and magnetic bending must also be considered before interpreting an endpoint as detector acceptance.

1.2 Normalized decay fraction through the tracker

For the conditional angular calculation, we use a transverse-vector decay distribution in the relativistic-daughter approximation, with uniform azimuth:

$$w(c_*) = \frac{3}{8}(1 + c_*^2), \quad \mathcal{A}(m, x) = \int_{-1}^1 dc_* w(c_*) \int_0^{2\pi} \frac{d\phi_*}{2\pi} \mathcal{I}_{\text{geom}}(p_+, p_-). \quad (7)$$

Here $\mathcal{I}_{\text{geom}}$ is one when both forward daughters satisfy the chosen geometrical requirements and zero otherwise. The integral is normalized to all decays with that specified polarization, parent energy and direction. Finite m_e is retained in the kinematics above; the adopted angular weight is not an exact threshold or production-polarization model.

For an active sensor rectangle centered at $\mathbf{r}_j$, let $\mathbf{n}_j$ be its normal, $\mathbf{a}_j, \mathbf{b}_j$ its orthogonal in-plane unit vectors, and $h_{a,j}, h_{b,j}$ its active half-widths. For the zero-field reference, a straight daughter ray $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{u}$, with $\mathbf{u} = \mathbf{p}/|\mathbf{p}|$, crosses the sensor plane at

$$t_j = \frac{(\mathbf{r}_j - \mathbf{r}_0) \cdot \mathbf{n}_j}{\mathbf{u} \cdot \mathbf{n}_j}, \quad \Delta \mathbf{r}_j = \mathbf{r}_0 + t_j \mathbf{u} - \mathbf{r}_j, \quad (8)$$

$$|\Delta \mathbf{r}_j \cdot \mathbf{a}_j| \leq h_{a,j}, \quad |\Delta \mathbf{r}_j \cdot \mathbf{b}_j| \leq h_{b,j}.$$

A geometrical sensor hit requires $t_j > 0$, a nonparallel ray, and both bounds. For station k in a specified detector half, define binary flags $h_{k,a}$ and $h_{k,s}$ for its axial and stereo views. Each flag is one if the ray crosses any active rectangle belonging to that view. Where hole and slot sensors are alternatives in one view, their rectangles form a union: crossing either is sufficient, and that view is counted at most once. The geometrical counts for one daughter are then

$$N_{2D} = \sum_k (h_{k,a} + h_{k,s}), \quad N_{3D} = \sum_k h_{k,a} h_{k,s}. \quad (9)$$

Thus a 2D plane hit requires one view, whereas a paired 3D space point requires both views at the same station and in the same detector half. These counts are not interchangeable when only one view is crossed: $N_{2D} = 2N_{3D}$ holds only if every crossed view has its partner. The run- and charge-dependent thresholds, including any required inner-plane hits, are specified in the selection table below. They enter $\mathcal{I}_{\text{geom}}$ separately for the electron and positron. Equation (8) gives the analytic crossing for the zero-field reference. With a magnetic field, the curved daughter trajectory determines the sensor crossings before the same view and

station flags are evaluated. These counts omit scattering and hit inefficiency.

1.2.1 Magnetic propagation

Let s denote path length, $q = +1$ for the positron and $q = -1$ for the electron, and $\mathbf{B}(\mathbf{r})$ the static field in the detector coordinate frame. For momentum in MeV, position in mm and field in tesla, the transport equations are

$$\frac{d\mathbf{r}}{ds} = \mathbf{u} = \frac{\mathbf{p}}{|\mathbf{p}|}, \quad \frac{d\mathbf{p}}{ds} = \kappa q \mathbf{u} \times \mathbf{B}(\mathbf{r}), \quad (10)$$

$$\kappa = 0.299792458 \frac{\text{MeV}}{\text{T mm}}.$$

The magnetic force is perpendicular to the momentum, so $d|\mathbf{p}|^2/ds = 0$: each daughter's momentum magnitude and energy remain constant in this transport model. The field changes the trajectory and the directions at the sensors. The mass labeling the acceptance curves is the invariant mass of the two momenta at the production vertex, rather than a mass recomputed from their different local directions downstream. Energy loss and material scattering require additional physics.

The numerical trajectory crosses sensor plane j at a forward path length $s_j > 0$ satisfying

$$[\mathbf{r}(s_j) - \mathbf{r}_j] \cdot \mathbf{n}_j = 0. \quad (11)$$

The in-plane half-width bounds of Eq. (8) are then tested at $\mathbf{r}(s_j)$. This prescription follows each charge through the same field and retains the complete finite sensor geometry. Setting $\mathbf{B} = 0$ recovers the straight-ray calculation. The field map, integration method and numerical checks are specified with the detector configuration below.

1.2.2 Projected angular reference

For an illustrative projection, consider symmetric active vertical intervals $y_{\min,j} \leq |y| \leq y_{\max,j}$ at longitudinal positions z_j , measured from a target at z_0 . A straight trajectory from the target has $y_j = (z_j - z_0)p_y/p_z$. Requiring each daughter to cross every chosen interval reduces to a constant slope band

$$L \equiv \max_j \frac{y_{\min,j}}{z_j - z_0} \leq \left| \frac{p_y}{p_z} \right| \leq U \equiv \min_j \frac{y_{\max,j}}{z_j - z_0}, \quad (12)$$

provided $0 \leq L < U$. This band is a reference for understanding the angular scales, rather than a replacement for the finite sensor calculation.

The azimuthal integral can be performed explicitly. At fixed c_* , set $p_T = p_* \sqrt{1 - c_*^2}$ and, for $p_T > 0$ and $p_{z,+}, p_{z,-} > 0$, define

$$q_{\text{low}} = \frac{L \max(p_{z,+}, p_{z,-})}{p_T}, \quad q_{\text{high}} = \frac{U \min(p_{z,+}, p_{z,-})}{p_T}. \quad (13)$$

Both daughters pass if $q_{\text{low}} \leq |\sin \phi_*| \leq q_{\text{high}}$. Writing $[q]_0^1 = \min(1, \max(0, q))$ and $[a]_+ = \max(0, a)$, the accepted azimuthal fraction and the remaining integral are

$$F_\phi(c_*) = \frac{2}{\pi} \left[\arcsin([q_{\text{high}}]_0^1) - \arcsin([q_{\text{low}}]_0^1) \right]_+, \quad \mathcal{A}_{\text{band}} = \int_{-1}^1 dc_* w(c_*) F_\phi(c_*). \quad (14)$$

We set $F_\phi = 0$ when either daughter is backward or the nonzero-gap aperture cannot be reached. This fraction describes geometry under the stated assumptions. A selected mass spectrum additionally depends on production, beam exposure, energy and direction distributions, trigger, reconstruction and analysis requirements; it is not a measurement of $\mathcal{A}(m, x)$.

1.3 A numerical boost example

For a 90 MeV parent carrying the 2016 beam energy, $E_A = 2.30$ GeV, Eq. (1) gives $\gamma = 25.556$ and $\beta = 0.999234$. At equal sharing, $E_+ = E_- = 1150$ MeV, $p_* = 44.997$ MeV and $p_{z,+} = p_{z,-} = 1149.119$ MeV. Each daughter is therefore at $\theta = 39.138$ mrad to the beam; the opening angle is 78.276 mrad. Rotating the decay azimuth changes its vertical projection while preserving the pair mass and polar opening. This is why a mass alone does not determine whether an event crosses the active silicon.

For comparison, the nominal 15 mrad vertical design angle gives, in the equal-sharing vertical plane at $x = 1$ and neglecting m_e ,

$$m_{15} \simeq E_{\text{beam}} \sin(0.015) = (15.84, 34.50, 56.10) \text{ MeV} \quad (15)$$

for 2015, 2016 and 2021, respectively. These are reference mass scales; the 15 mrad gap alone provides no upper mass cutoff.

2 Detector geometry, fields and hit selection

The beam energies are 1.056, 2.30 and 3.74 GeV for 2015, 2016 and 2021, respectively [1, 2, 3]. The detector input is the committed LCDD geometry in `JeffersonLab/hps-java`, fixed to revision 473c732d [4]; the complete identifier is saved in the source manifest. LCDD contains the active sensor solids and their complete placement hierarchy. The corresponding compact XML and Java geometry definitions are retained as supporting sources; a fresh rebuild of those definitions is not assumed to reproduce the committed alignment exactly.

The selected configurations are `HPS-EngRun2015-Nominal-v6-0-fieldmap`, `HPS-PhysicsRun2016-Pass2` and `HPS_Run2021Pass2FEE`. The first is a representative nominal engineering-run configuration. The second appears in local HPSTR radiative-MC filenames. The third is explicitly selected by the local 2021 HPSTR simulation job list, with run 14166 and target $z = 0$. Exact alignment and selection equivalence to the three observed histograms has not been established.

The extracted geometries contain 36, 36 and 40 active silicon rectangles. Standard sensors have active dimensions $38.3399 \times 98.33 \times 0.32$ mm. The first two 2021 stations instead use $14.025 \times 30 \times 0.20$ mm sensors. There are six station pairs in the engineering-run layouts and seven in 2021. In the 2021 file, the first station corresponds to the added physical L0; its axial and stereo planes are staggered, as are the top and bottom planes. The earliest sensor centers lie near $z = 38$ mm in 2021 and $z = 88$ mm in the older layouts. A nominal station distance of 50 or 100 mm therefore does not specify every active-plane position.

2.1 Coordinates and hit requirements

The parent boost in Section 1 is defined in the beam frame. The compact geometries specify a beam angle $\alpha = 0.03052$ rad. We rotate the daughter momenta into the LCDD frame using the sign convention of `hps-mc` [5]:

$$\begin{pmatrix} p_X \\ p_Y \\ p_Z \end{pmatrix} = \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix} \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix}. \quad (16)$$

The parent direction is consequently $(\sin \alpha, 0, \cos \alpha)$. The model vertex is $(0, 0, 0)$ mm. No physical target solid was found in these LCDD snapshots; the origin is a stated nominal-vertex assumption, with only the 2021 target- z choice directly supported by the cited local job list.

The main curves apply the run-dependent hit requirements in Table 1, using the geometrical hit counts defined in Eq. (9). A paired 3D point requires an axial and a stereo crossing in the same detector half. A 2D count requires only the individual view. The hole and slot sensors within one view remain alternatives.

Run	Positron	Electron	Selection provenance
2015	$N_{3D} \geq 5$ of 6; paired hits in L1 and L2	$N_{3D} \geq 5$ of 6	Internal note and local v5.0.4 convention [7, 8]
2016	$N_{3D} \geq 5$ of 6	$N_{3D} \geq 5$ of 6	Published 2016 selection, Sec. III.2.2 [3]
2021	$N_{2D} \geq 10$ of 14	$N_{2D} \geq 8$ of 14	Current analysis specification supplied for this study

Table 1: Hit requirements implemented for the main geometric curves. The 2015 inner-station requirement is charge dependent; no additional mandatory inner station is imposed for 2016 or 2021. The 2021 counts are individual 2D plane hits, not paired 3D points.

For 2015, the internal note specifies five of six tracking points on PDF page 11 and the positron inner-layer condition in Table 3 on PDF page 12; the local v5.0.4 selection follows this convention. The published 2015 search instead required both tracks to have hits in the first two layers [2]. The present comparison therefore follows the stated local-analysis convention. The 2021 row adopts the current specified hit counts, superseding the older 2021 statements in the v5.0.4 snapshot. Equivalence between this row and the selections of the saved 2021 data histogram has not been established.

Solid curves use these selected-hit requirements with magnetic transport at $x = 1$; dashed curves use the same requirements at $x = 0.8$. Grey dotted curves give the corresponding $x = 1$ zero-field calculation. A faint dash-dotted reference requires both views in every station with the field present: all six stations for 2015/2016 or all seven for 2021.

2.2 Field maps and numerical propagation

The field tables are taken from `JeffersonLab/hps-fieldmaps` at revision `feadfcdb`, with the full identifiers and file hashes saved in the input manifest [9]. The 2015 calculation adopts the corrected-unfolding v3 map with scale 0.7992. Its compact detector definition has the same non-field geometry content as the original configuration, but no separately generated v3 LCDD is provided; the original committed sensor transforms are therefore retained. The 2016 calculation uses the scale-1.04545 v4 map explicitly referenced by the selected Pass2 detector.

For 2021, the scale-1.07326 map is documented for physics data at 3.742 GeV. We combine it with the retained Pass2FEE sensor geometry and use the nominal rounded beam energy of 3.74 GeV. This is an explicit combination of a physics-data field and the selected sensor snapshot. It is not an assertion that the Pass2FEE alignment equals the data configuration: the same sensor names and sizes occur in the data-map detector, but their positions and orientations differ. Exact alignment and histogram-selection equivalence remain unverified.

All three maps use 5 mm grid spacing and the world offset $\mathbf{r}_{\text{off}} = (21.17, 0, 457.2)$ mm. The map is queried at $\mathbf{r} - \mathbf{r}_{\text{off}}$, and each component is interpolated trilinearly. The raw field values are multiplied by 1000 to obtain tesla, as verified against both the LCDD/SLIC and Java field readers [10]. The table components already refer to the detector axes; they receive no additional beam rotation or sign change. The central B_Y values at the map origin are approximately -0.240245 , -0.5234 and -0.8596 T, respectively. The map origin is not the target.

Equation (10) is integrated with fourth-order Runge–Kutta steps in path length. The maximum step is 5 mm, reduced when necessary to limit the estimated angular change to 0.01 rad:

$$h \leq \min\left(5 \text{ mm}, \frac{0.01 |\mathbf{p}|}{\kappa B_{\text{max}}}\right). \quad (17)$$

Sensor crossings are located on a cubic Hermite interpolation of each step, with a normal-plane residual

tolerance of 10^{-7} mm before the finite-face bounds are applied. A zero field is used outside the tabulated grid, matching the source readers. All active sensor corners lie inside the transverse map bounds and upstream of the stopping plane $Z = 940$ mm. Propagation stops at that plane, at the first nonpositive u_Z , or on an outward transverse exit from the grid. Thus the calculation follows forward tracks; it does not include later crossings by trajectories that turn back in Z .

2.3 Zero-field projected mass scales

For comparison with the nominal angular picture, each active rectangle can be projected onto vertical position Y and distance along the beam $\ell = \mathbf{r} \cdot (\sin \alpha, 0, \cos \alpha)$. The extrema of Y/ℓ over its four corners define a zero-field projected slope interval. The same sensor unions and the selected-hit criteria in Table 1 are then applied in this projection to symmetric upper and lower rays. Both charge assignments are tested, and the accepted intervals are combined by union. Table 2 converts the resulting angular interval into an equal-sharing mass scale using Eq. (6) at $x = 1$.

Run	E_{beam} [GeV]	Stations	θ_{low} [mrad]	θ_{high} [mrad]	m_{vert} [MeV]
2015	1.056	6	16.33	68.46	17.27–72.24
2016	2.300	6	16.32	68.53	37.54–157.50
2021	3.740	7	14.57	93.32	54.50–348.53

Table 2: Zero-field projected vertical aperture of the selected LCDD geometry with the run-dependent hit requirements of Table 1. The angular scan spacing is 0.002 mrad. These are equal-sharing mass scales in a two-dimensional projection. They are not the mass endpoints of the full geometry with magnetic transport.

The projected intervals are approximately 17.27–72.24 MeV for 2015, 37.54–157.50 MeV for 2016 and 54.50–348.53 MeV for 2021. Their angles and mass scales depend on finite plane offsets, staggering and the specified hit counts. In particular, the 2021 requirement permits missed individual views; it does not force a hit in every station. A zero-field ray at an interval edge can miss some sensor faces while satisfying the projected hit criterion. These scales remain a separate angular reference; the following display uses accepted curved trajectories from the full calculation.

Projection also removes horizontal gaps. This is a substantive limitation, not merely a drawing convention. For example, a zero-field ray in the exact vertical beam plane runs near $X = 15$ mm at the fourth 2015 station, between the axial hole sensor ending near 13.7 mm and the slot sensor starting near 16.3 mm. Such a ray can lie inside the projected vertical interval while missing active silicon. The full three-dimensional calculation retains this seam and propagates each charge through the field before testing a crossing.

2.4 Sampled magnetic response

The selected-hit magnetic curves have sampled maxima at 40, 90 and 195 MeV, with fractions approximately 26.09%, 25.95% and 40.11%, respectively. Table 3 records the first and last sampled masses with a passing orientation. The examples in the following display are chosen from those passing events. A finite mass grid and a finite set of orientations cannot establish a physical acceptance endpoint.

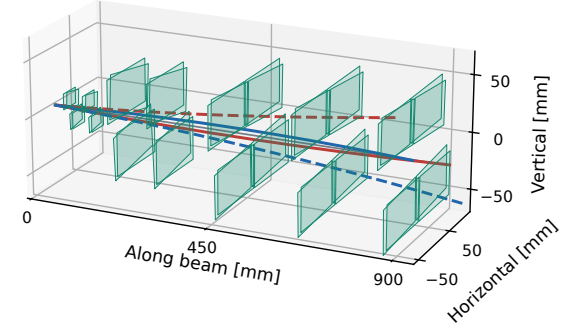
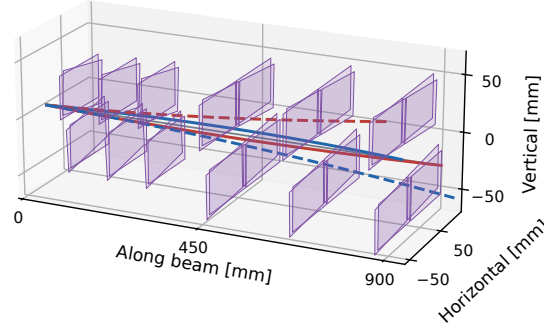
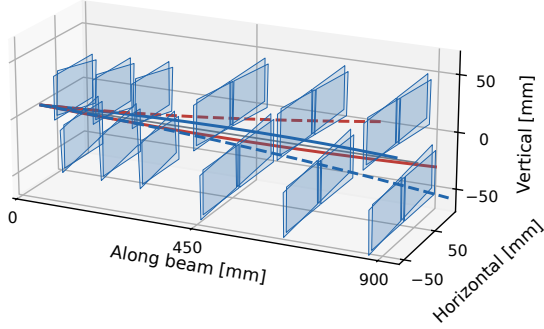
3 Detector geometry and example trajectories

HPS magnetic transport through the active silicon tracker

HPS 2015 | 1.056 GeV

HPS 2016 | 2.300 GeV

HPS 2021 | 3.740 GeV



Passing examples: 20 and 195 MeV

Passing examples: 40 and 425 MeV

Passing examples: 60 and 715 MeV

Nominal 15 mrad scale ≈ 15.84 MeV

Nominal 15 mrad scale ≈ 34.50 MeV

Nominal 15 mrad scale ≈ 56.10 MeV

— Positron — Electron — Low-mass example - - - High-mass example

Examples pass the run-dependent hit cuts at the first and last nonzero sampled masses; they do not define exact endpoints.
Pinned 3D field maps; transverse display scales expanded. 15 mrad labels use equal sharing, $x = 1$ and negligible electron mass.

Figure 1: Standalone three-column display of active sensor faces and accepted electron–positron trajectories in the field. Each panel shows examples at the first and last mass with a nonzero selected fraction on the saved mass and orientation grid. These are passing examples, not physical acceptance endpoints. The nominal 15 mrad equal-sharing scale is shown as an angular reference. Sensor staggering and horizontal gaps enter the full three-dimensional crossing test; a passing track need not hit every station. Transverse display scales are expanded for clarity. The detector and field configurations are those specified above. The display is supplied separately as vector PDF/SVG and PNG.

Run	$B_y(\mathbf{r}_{\text{map},0})$ [T]	Sampled masses [MeV]	Peak [MeV]	Peak fraction
2015	-0.2402	20–195	40	26.1%
2016	-0.5234	40–425	90	25.9%
2021	-0.8596	60–715	195	40.1%

Table 3: Magnetic response at $x = 1$ with the run-dependent hit criteria. The field is quoted at the map origin. The nonzero sampled-mass interval and peak refer to the saved 5 MeV mass grid with 4096 decay orientations; they are not exact kinematic limits or a calibrated efficiency.

4 Acceptance curves and selected mass spectra

The upper row of Fig. 2 integrates the finite-sensor test after magnetic transport over the daughter decay orientation. The calculation uses 4096 deterministic Sobol points at each mass, with 5 MeV spacing. The panels extend to 250, 550 and 950 MeV for 2015, 2016 and 2021, respectively. The zero-field comparison uses the same hit requirements and parent energy.

The all-station reference is more restrictive. The $x = 0.8$ curves show how the response changes when the same mass has a smaller boost. Events outside the projected mass interval can still cross the full geometry at other azimuths or with unequal sharing. Neither the interval in Table 2 nor the first or last nonzero sampled mass should be used as an absolute acceptance threshold. The finite mass grid also does not prove zero support between or beyond sampled points.

The bottom row retains the original selected-pair spectra in 1 MeV bins: 2015 full, 2016 full and 2021 10%. Their displayed integrals are 21,442,838, 73,218,251 and 141,305,897 pairs. The display starts at 15, 30 and 36 MeV, respectively, and ends at 300 MeV. The 2015 input ends at 150 MeV; its grey region denotes unavailable bins. The samples have no exposure or luminosity rescaling, and their selections differ. Their shapes therefore provide observational context, not a validation of the signal geometry fractions or an absolute production rate.

4.1 Validation and scope

The calculation now uses the sensor sizes, orientations, staggering, horizontal seams and beam rotation encoded by the detector files, together with the specified magnetic fields and hit criteria. It preserves exact two-body kinematics, including the electron mass, while using the specified relativistic transverse-vector angular law. Geometry transforms and the boost were checked independently. Both charge signs reproduce analytic circular motion in a uniform field; zero-field transport reproduces the direct ray-intersection hit counts for 1024 test orientations per year. Using steps of 5, 2.5 and 1.25 mm gives identical selected-event classifications for 1024 orientations at one representative mass per year: 58.08, 126.5 and 205.7 MeV. Doubling the orientation sample from 4096 to 8192 at three masses per year changes the selected-hit or all-station fractions by at most 0.002371, less than 0.238 percentage point. These are sampled numerical checks, not a bound on the error at every mass or a detector-model uncertainty. Count-preserving histogram checks also passed.

Material interactions, hit-finding efficiency, dead channels, beam spot, the parent production spectrum and angular distribution, trigger, reconstruction, other kinematic and analysis cuts, and displaced decays remain outside this study. In particular, no integration over the produced A' energy or direction distribution is performed. The result is a conditional geometry-and-field acceptance for the stated hit requirements, not a calibrated HPS signal efficiency or a prediction of accepted yield.

Magnetic-field acceptance and selected mass spectra

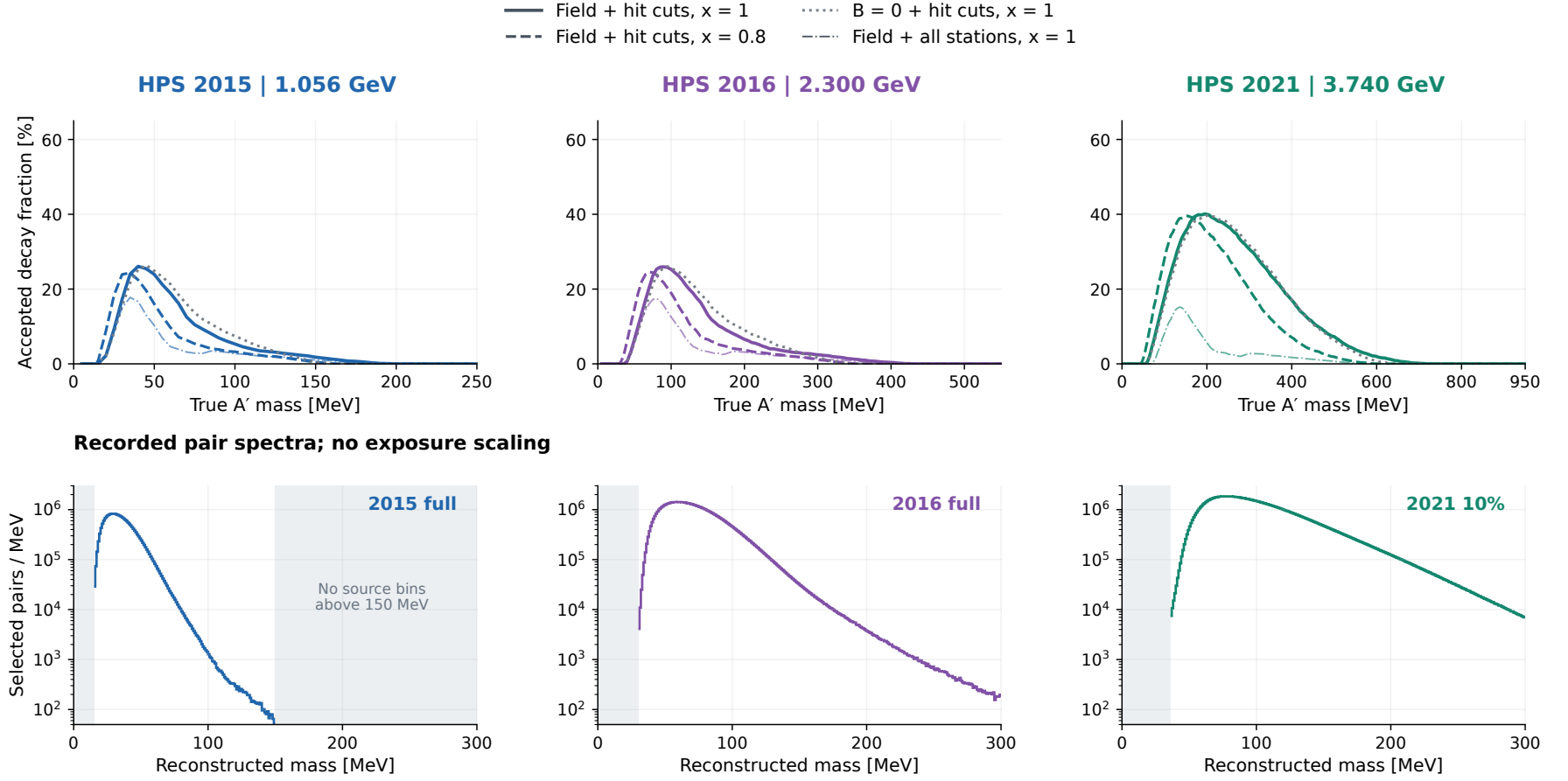


Figure 2: Both-daughter fractions after transport through the active sensor geometry (upper panels) and raw selected-pair densities (lower panels). Solid and dashed curves apply the run-dependent hit criteria in Table 1 with the field present at $x = 1$ and $x = 0.8$, respectively. Grey dotted curves give the selected-hit zero-field reference at $x = 1$; faint dash-dotted curves require all station pairs with the field present at $x = 1$. The parent energy is $E_A = xE_{\text{beam}}$. These panels and Fig. 1 are also supplied together as the requested standalone 3×3 overview. They are separated here to preserve label legibility on the page.

References

- [1] N. Baltzell et al., *The Heavy Photon Search Experiment*, [arXiv:2203.08324](#) (2022), Table I and detector description.
- [2] P. H. Adrian et al., *Search for a dark photon in electroproduced e^+e^- pairs with HPS*, Phys. Rev. D **98**, 091101 (2018), [arXiv:1807.11530](#).
- [3] P. H. Adrian et al., *Searching for prompt and long-lived dark photons with HPS*, Phys. Rev. D **108**, 012015 (2023), [arXiv:2212.10629](#).
- [4] Jefferson Lab, *hps-java detector descriptions*, revision 473c732d, [detector-data/detectors](#). Committed compact and LCDD snapshots and their SHA-256 hashes are included in this package.
- [5] Jefferson Lab, *hps-mc beam coordinate conversion*, revision 308e27be, [beam_coords.cc](#).
- [6] M. Battaglieri et al., *The Heavy Photon Search Test Detector*, [arXiv:1406.6115](#). The historical approximate 15–70 mrad design interval motivated the original v5.7.0 illustration; it is not used as the finite-geometry boundary in v5.7.1.
- [7] O. Moreno, N. Baltzell, M. Graham and J. Jaros, *Search for a Heavy Photon in Electro-Produced e^+e^- Pairs with the HPS Experiment at JLab*, archived HPS internal note (undated local copy), PDF pp. 11–12, Table 3. The local file hash and source locators are recorded in `inputs/selection/selection_criteria.json`.
- [8] E. Peets, M. Gignac and E. Hsu, *HPS Gaussian-Process Resonance Search: Analysis Note v5.0.4*, local analysis-note snapshot dated 11 September 2026, event-selection section and table. Unmodified source excerpts are included in `inputs/selection/`; the current user correction supersedes their older 2021 hit-count statements for this study.
- [9] Jefferson Lab, *HPS magnetic-field maps*, revision feadfcdb, [run-period catalogue](#). Active archives: [2015 corrected v3](#), [2016 v4](#), [2021 data map](#). Archive and extracted-file hashes are recorded with the active field configuration.
- [10] Jefferson Lab, *hps-lcsim field-map reader*, revision 391d4649, [FieldMap3D.java](#); SLAC, *LCDD Cartesian field-map reader*, revision 36a4805e, [Cartesian3DMagneticFieldMap.cc](#). These sources establish field units, global-offset subtraction and interpolation.